

Virtual Technical Conference Platform

Use-Case Flow Specification: Generate Schedule (UC-05)

Actor(s): Track Chair (primary); Conference Author (secondary — availability constraint)

Related Requirements: FR-002 (conflict-free scheduling), NFR-001 (10-second resolution)

Preconditions

- All papers for the current cycle have completed review and been marked Accepted or Rejected.
- Each accepted paper is assigned to a track and has a confirmed presenting Author.
- Room capacities and time-slot availability for the venue have been entered into the system.
- Authors have submitted their availability constraints (blackout times) for the conference dates.

Postconditions

- A complete multi-track schedule exists with every accepted paper assigned a room, time slot, and track.
- No two sessions share the same room and time slot, and no Author/speaker is double-booked.
- The schedule is visible to Track Chairs and to Authors for their own sessions.

Main Success Scenario

1. Track Chair selects "Generate Schedule" from the dashboard.
2. System retrieves the list of accepted papers, their tracks, and assigned presenting Authors.
3. System retrieves room capacities and available time slots for the conference venue.
4. System retrieves each Author's submitted availability constraints.
5. System runs the scheduling algorithm, assigning each session a track, room, and time slot while satisfying all constraints.
6. System checks the generated schedule for conflicts (room double-booking, speaker double-booking, capacity overflow).
7. System confirms zero conflicts and displays the draft schedule to the Track Chair within 10 seconds (NFR-001).
8. Track Chair reviews and publishes the schedule.
9. System notifies all Authors of their assigned session time, room, and track.
10. Use case ends successfully.

Alternate Flow

6a. Conflict Detected (system cannot fully resolve constraints)

- 6a1. System flags the specific unresolved conflict (e.g., an Author with overlapping availability blackouts across two accepted sessions).
- 6a2. System displays the conflict to the Track Chair with the affected sessions and Author(s) highlighted, instead of a completed schedule.
- 6a3. Track Chair manually reassigns the affected session to a different time slot or room.
- 6a4. System re-runs the conflict check on the adjusted schedule.
- 6a5. If no conflicts remain, flow resumes at step 7 of the Main Success Scenario. If conflicts persist after 2 manual adjustment attempts, the Track Chair is prompted to contact the affected Author(s) directly, and the use case ends without a published schedule.